

# Automatic Recognition of Rape Seeding emergence Stage based on Computer Vision Technology

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**Abstract**—Acquisition of crop growth stage information can not only help to analyze the relation between the crop growth process and environmental condition, but also to guide the field operation effectively. Therefore, different growth stages of rape crops are monitored with the visual system constructed in this paper, and the first critical growth stage of rape is detected automatically, which is seeding emergence stage. The rape should be first extracted from the image. Considering the impacts of the complicated environment and climatic changes, HI color segmentation method is adopted to segment the crops from the background. Then, two limited conditions, cotyledon area and density, are applied to judge whether it is at seeding emergence stage. Eventually, the experimental results are compared to the ones from other mature methodologies and manual observation, and it shows that the proposed methodology is effective and feasible, and it can provide support for precision agriculture.

**Keywords**—Rape; HI; seeding emergence stage; automatic detection technology; precision agriculture

## I. INTRODUCTION

As one of the most important technology in agricultural development, digital agricultural technology has integrated modern information technology and artificial intelligence technology into agriculture. As for the crop growth monitoring, digital agriculture is required to achieve the information about key growing period of plants accurately in real time, which may guide the delicacy management of plant growing stage. The monitoring of crops in different stages is a significant part in the plant growth monitoring. At first, the understanding about the plant growth stage may help people analyze the relationship between crop growth process and environmental conditions. In addition, it can also guide the field operation effectively, such as thinning, filling the gaps with seedlings, fertilization, irrigation, trimming, and insect disease prevention. In this way, the crop yield will be improved evidently [1,2].

Monitoring of crop growth and development at home and abroad mainly includes remote sensing monitoring, plant growing model simulation and ground observation, etc. Although these methods obtained some achievements in the crop growth monitoring, they may be characterized by waste of time and energy, huge error and poor instantaneity, etc. in

acquiring information. The organic integration of high-tech and modern agriculture brings new power to the development of agriculture, which may improve the timeliness and accuracy of crop growth monitoring greatly. Computer vision technology has been widely researched and applied in the non-destructive monitoring of crop growth stage. Compared with other monitoring method, it is characterized by substantial data collection, rapid speed, high precision, etc. Furthermore, it has huge potential in saving labor and lowering the subjectivity of human judgment [3,4]. Till now, images acquired with visual system are employed for conducting automatic monitoring of crop growth, and substantial research work has been conducted, which can be concluded in the following aspects: (1) detection and recognition of disease, insect pests and weeds [5]; (2) measurement of external crop growth parameter (such as blade area, height, etc.) [6]; (3) lack of estimation for crop nutrition [7]; (4) recognition and categorization of crops. All the above work focuses on the study on specific organs of crops, such as leaf, flower, etc. Few studies involve the automatic recognition of crop growth stage with computer vision system [8,9]. Furthermore, researchers, like Zhenghong Yu et al. conducted pioneering researches in this aspect, and recognized the emergence stage and three-leaf stage of corns automatically with computer vision technology [10].

Rape is one of the most significant crops in China, as well as the most significant raw materials of fodder, chemical engineering and energy. Therefore, it has great strategic and economic significance in national economic production. In recent years, the rape production is severely impacted, and its economic benefit is quite low. Consequently, it is quite necessary and urgent to conduct automatic recognition and detection for the growth of rape with the assistance of computer vision technology. In this paper, rape photos under different climatic conditions are obtained with cameras successively. Meanwhile, rape is extracted via color segmentation algorithm of HI, which is flexible to the changes of different environments, and the segmentation result is compared with four other segmentation results. On the basis of segmentation results, two limited conditions (cotyledon area and density) of rape in seedling stage are applied. Eventually, the recognition results are compared with the results of manual observation.

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## II. MATERIAL AND METHOD

### A. Experimental field and crop

The Experimental field is located in Huazhong Agricultural University (Wuhan, Hubei), and rape type is Huayouza 62, planting pattern is directly seeding.

### B. Image acquisition

Pictures of rape are taken in different time periods within 35 consecutive days with a Canon camera.

### C. Color segmentation of crop

The primary work of recognizing the seedling emergence stage of rape is to correctly segment the pixel point containing rape, which may impact the follow-up work. At present, researchers have already proposed some decisive measures for recognizing the crops in image, especially for the segmentation of green crops. It mainly consists of the following types: 1. visible spectral-index based, including the excess green (EXG) [11], the excess red (EXR) [12], the excess green minus excess red index (EXGR) [13], the color index of vegetable (CIVE) [14] and Vegetative (VEG) [15]; but all these measures require a fixed threshold value for the final segmentation; 2. specific threshold-based approaches; the premise of applying this approach is to assume two classification problems (classification of plants and soil). Kirk et al. (2009) applied a combination of greenness and intensity derived from the red and green spectral bands to divide all pixels into two types, including the plants and soil [16]. Meyer and Camargo-Neto (2008) applied the automatic OTSU threshold value method to segment the binarized EXG and normalized difference index (NDI) [17]. There will be serious error segmentation if the light conditions change evidently with time and weather factor. 3. Learning-based; Tian and Slaughter (1998) proposed the recognition of crops with EASA through the process of monitoring the study [18]. Then, Ruiz-Ruiz et al. (2009) dealt with the influence of illumination variations by combining with the HSI color space on the ESAS basis [19]. Zheng proposed the mean-shift green plant index approach by training with neural network and by taking advantage of the green component of green plants with high component value [20]. It must input substantial training samples, and it can achieve good segmentation results only with the assistance of artificial interaction. However, the HI color segmentation algorithm only requires small training sample, and it shows great robustness to different environmental changes; thus it satisfies the extraction of green crops in the great context.

#### 1) HI color model

Since the color of single-color object distributed on the hue-saturation(HS) plane changes with luminance [21], the green pixel model applied in HSI color space shall adapt to the changes in external light condition. However, according to the image statistics of crops, the histogram of hue is similar to Gaussian distribution in certain strength (I). Therefore, with given strength I, the hue of green H follows Gaussian distribution. The probability density function is given in equation (1), where h (hue) stands for the green hue,  $\mu$  is the

expectation, and  $\sigma^2$  is the variance. Maximum likelihood method can be applied for evaluating the sample data set to

work out the distribution parameter  $\hat{\mu}$  and  $\hat{\sigma}^2$  of each type of strength, and the solving method is displayed in equation (2). In the experiment, 20 outdoor rape pictures in different growth stages in different weather conditions are taken as the sample data, and then pictures with green information besides the background is achieved (Fig.1) through manual setting of color channel threshold value. Later, the green information of sample data is transformed into the HSI color space model from RGB color space model. Finally, hue-intensity look-up table (HI-LUT) is established for the green information. Each item in the look-up table includes  $\mu$  and  $\sigma^2$ , and the strength ranges in [1, 255]

$$f_H(h|I) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2\sigma^2}(h-\mu)^2\right] \quad (1)$$

$$\hat{\mu} = \frac{\sum_{i=1}^n H_i}{n}, \quad \hat{\sigma}^2 = \frac{\sum_{i=1}^n (H_i - \hat{\mu})^2}{n} \quad (2)$$

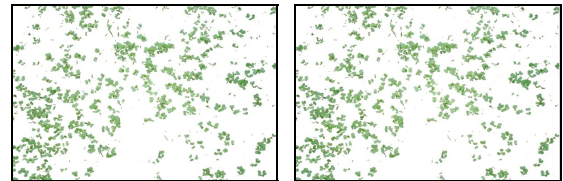
According to HI color model, if  $\psi(i, j)$  represents the distance between the hue of a given pixel  $(i, j)$  ( $(i, j)$  denotes the coordinate of pixel in the image) and the expected hue. The greater the distance is, the smaller the possibility of the pixel point being the green crops will be. In equation (3),  $H(i, j)$  and  $I(i, j)$  signify the huge and intensity of the coordinate  $(i, j)$  respectively,  $\mu_1(i, j)$  and  $\sigma_1(i, j)$  are from the average value and stand derivation of the look-up table respectively. According to this result, an appropriate threshold value can be set easily for image segmentation, as shown in equation (4). When  $\psi(i, j)$  is smaller or equals k, the pixel belongs to the crop, or it belongs to the background.

$$\psi(i, j) = \frac{|H(i, j) - \mu_1(i, j)|}{\sigma_1(i, j)}$$

$$pixel(i, j) \in crop; \quad \text{if } \psi(i, j) \leq k \quad (3)$$

$$pixel(i, j) \in background; \quad \text{else} \quad (4)$$

However, how to select an appropriate threshold value k is a problem that shall be solved, for different threshold value k may give rise to different results. But if the threshold value is too large, some non-color pixels may be included in the green pixel. Therefore, the selection of threshold value k will affect the accuracy of crop segmentation greatly.



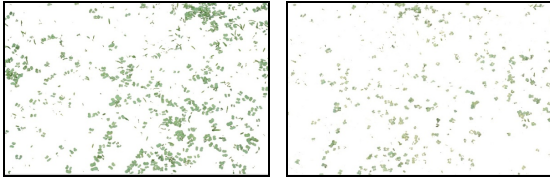


Fig. 1. Rape sample under different climatic conditions

#### D. Automatic detection method in seedling emergence stage

The seedling emergence stage is the first crucial stage in the entire growth stage of rape. When crops reach this stage, farmers must detect if thinning and supplementing are required. Timely thinning and supplementing can guarantee that crops are in proper planting density. Furthermore, proper planning density will not only allow crops to acquire more fertilizer, sunshine and space for growth, but also guarantee the crop yield and increase the production efficiency. Therefore, by combining with the practical condition of rape plantation, an approach based on the computer vision is proposed to automatically recognize seedling emergence stage of rape. At first, rape is extracted from the entire image, and then limiting conditions are added for confirming if the rape reaches seedling emergence stage.

Recognition of seedling emergence stage : at first, rape is extracted from the background with HI color segmentation algorithm. However, we find that there are monocotyledon weeds in sharp strip shape of the extracted rape, and rape has a unique characteristic in seedling period: it has two pieces of cotyledon in kidney shape, and the cotyledon distributes symmetrically in space. According to different shape features and areas, the limiting condition is set to eliminate weed, and the rape in seedling stage is recognized automatically.

As for the detection of seedling emergence stage, the leaf area is a feasible feature. According to the statistics about cotyledonal area in seedling emergence period, the cotyledonal area generally ranges between  $0.2\text{cm}^2$  and  $0.5\text{cm}^2$ . Determination for the cotyledonal area in the segmentation image shall focus on the practice from the pixel area in the image. There are mainly three steps: firstly, the 8-field contour tracing is conducted for the segmented binary image, so as to gain outline of the target. Later, the size of pixel point shall be calculated according to the ruler method, and then the practical area of pixel can be worked out. The ultimate area of target object equals the sum of the pixel area. Therefore, we can confirm if it is in the range of cotyledon area.

In order to automatically recognize whether rape is in seedling emergence stage, besides the precious feature, another new feature, namely the density is added. Density is the most common approach of measuring the circularity, and it is mainly applied for depicting the complexity of the boundary. The algorithm of circularity is the proportion of the square of the target perimeter and area. According to the statistical calculation of cotyledon density in seedling stage, there are obvious distinctions from other target objects (such as weeds, etc.). The density of cotyledon ranges between 15 and 19. Therefore, by setting the threshold values of area and density, rape in seedling emergence stage can be recognized.

### III. Result and discussion

#### A. HI color segmentation algorithm experiment

In order to verify the performance of HI color segmentation algorithm, 30 rape pictures in different weather conditions (sunny days, rainy days and cloudy days), complex environments, and different growth stages are selected from the overall data; and then segmentation is conducted with these pictures. By adjusting the value of  $k$ , the result of image segmentation shows that HI color model has a strong adaptability in different light conditions when  $k=2.5$ (Fig.2).. When  $k=1$ , there is a serious lack of rape, and it may treat some of the rape as the background by mistake. When  $k=4$ , the non-green pixel is divided as the green rape. In order to make  $k$  value adapt to the correct segmentation of the experimental data,  $k$  is adjusted again. We find that it may gain the best segmentation result when  $k$  ranges from 2.4 to 2.6, which is in accordance with the requirements of the follow-up research.

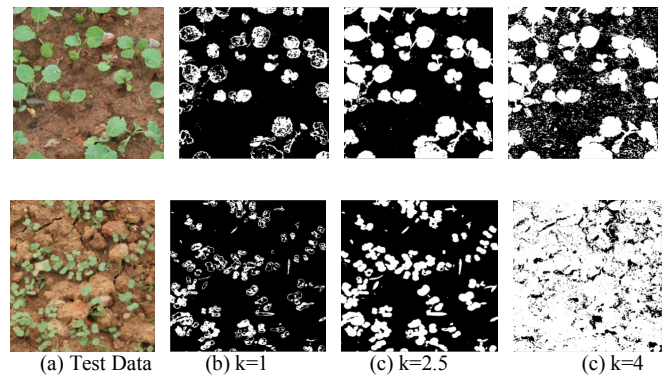


Fig. 2. HI segmentation result of different  $k$  values

After confirming that the HI color segmentation algorithm threshold value  $k$  can achieve perfect green rape, it is compared with other four approaches of extracting green crops (CIVE, EXGR, EXG and VEG). In these approaches, CIVE, EXGR and EXG require OTSU to achieve the threshold value; as for VEG, the binarized segmentation can be achieved by setting the mean value as the threshold value. Fig. (3) and Fig. (4) are the effect pictures of rape in two different weather conditions among the five different approaches, and it has been discovered that compared with other effect, HI color segmentation algorithm reaches the best effect. The severe loss of rape may be affected by distinct illumination intensity. For instance, with VEG method, there are missing phenomena in Fig (3), but there is a serious noise in Fig. (4). Besides, most the green pixel points are divided as green pixel points. EXG performs excellently in adapting to different weather effects, but compared with HI mode, there is still a small deficiency.

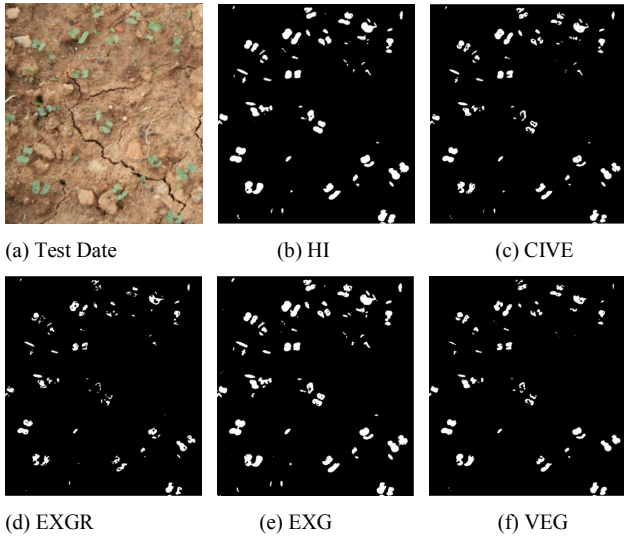
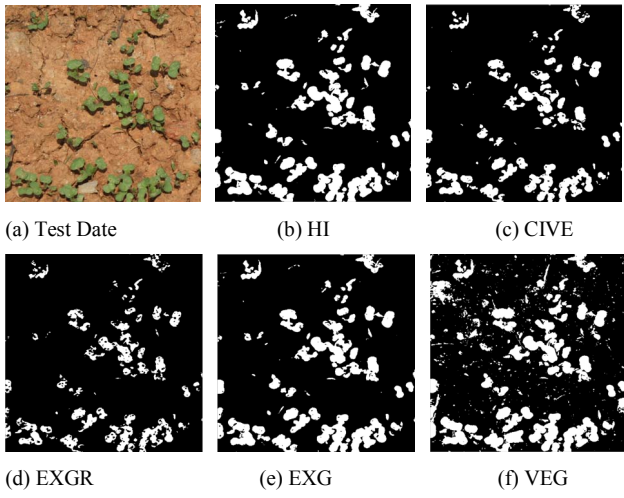


Fig. 3. Segmentation images of different segmenting measures



Segmentation images of different segmenting measures

### B. Seedling emergence stage recognition experiment

Fig. 3 (a) is the original picture in which we have to recognize whether the rape is in seedling emergence stage, and we see it contains both rape and monocotyledon weed. With HI color segmentation approach, when the threshold value  $k=2.5$ , Fig. 3 (a) will be segmented, and the segmentation result (Fig. 5 (a)) shows good robustness and maintains the basic form of green plants when compared with other segmentation results. However, other approaches present serious deficiencies and noise in the green plants. If these methods are used to recognize whether rape is in seedling stage, the recognition effect may be decreased severely. According to observation about the original picture, there are severe missing parts in cotyledon, which may be displayed after HI color segmentation (Fig. 5 (a)). Moreover, some missing parts may affect the calculation of area. Therefore, according to the morphological processing, it shall achieve a complete target object (Fig 5(b)). On that basis, the target contour can be extracted with the 8-field contour tracing method, and the cotyledon and rape are still

connected. They are separated gain with morphological method, and the final binary image is gained after a series of processing (Fig. 5 (c)).

Fig. 5 (d) is the 8-field contour tracing of Fig 5 (c). By setting the threshold value ( $[0.2\text{cm}^2, 0.5\text{cm}^2]$ ) of area, object satisfying the specific conditions is extracted. Some weeds and noises are eliminated. Fig. 5(e) is a figure obtained according to the settings of area threshold value. Later, on the basis of Fig. 5 (e), the threshold shall stick to the value limit, and it shall be limited by the density [15, 19]. Finally, it will achieve the rape in seedling emergence stage, as shown in Fig. 5 (f). Eventually, the final results (Fig 5 (f)) are compared with the manual observation results, and the results are the consistent.

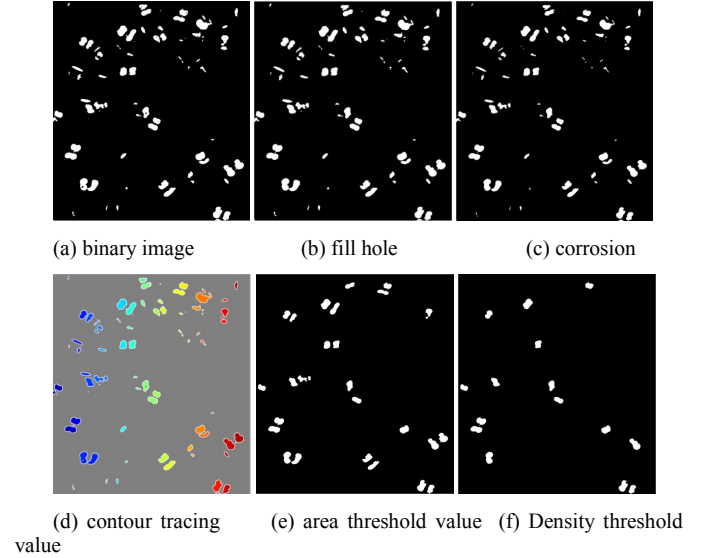


Fig. 4. Flow chart of recognizing the seedling emergence stage

### C. Deficiencies

HI color segmentation algorithm is applied, which requires the  $k$  value to regulate the segmentation of crops. Although we have determined an appropriate threshold value  $k$  [2.4, 2.6] for segmentation, some of the non-green pixels are still divided as the green pixels.

## IV. CONCLUSION

A new segmentation algorithm (HI) is utilized to extract rape from the image, and on that basis, the rape in seedling emergence stage can be recognized automatically. Such result is inspiring and can satisfy the actual demands. Meanwhile, it also shows that computer visual technology has huge potential in automatic detection of crop growth stage. The future research work: to improve the segmentation algorithm by adding clustering method on the basis of HI segmentation algorithm, and to reduce the impact of treating the non-green pixel point as the green pixel by mistake. Owing to the overlapping occurrence of rape in seedling emergence stage, the seedling emergence period in overlapping condition is also recognized automatically. Meanwhile, the automatic recognition of three-leaf and five-leaf rape will be launched,



for the three-leaf stage and five-leaf stage require final singling. Key growth stages of crops should be accurately and automatically recognized, and information about crop growth must be acquired and provided for farmers. This is the internal requirements of developing agriculture and digital agriculture, and it will certainly improve the labor efficiency and increase the yield.

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